Multiple Recovery Paths in a Lesioned Transformer

Internal Rerouting and Readout Remapping

Lorenzo Cannata (DISI) - Uri Hasson (CIMeC)

Department of Information Engineering and Computer Science - Center for Mind/Brain Sciences (University of Trento)

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Motivation

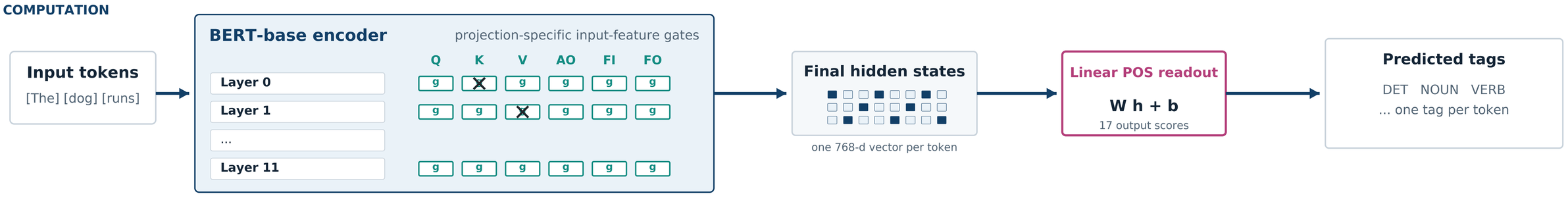
Using BERT as a frozen encoder and a linear part-of-speech (POS) readout as a task classifier, we first use gate masking to learn a sparse core subnetwork sufficient for POS classification. We then ablate this subnetwork to produce a large drop in performance. Our main question is whether performance can recover by adapting the surviving projection-specific gates, retraining the linear POS readout, or both.

CENTRAL FINDING

Recovery can use internal rerouting or readout retraining; among fixed-weight routes, combining both recovers most without necessarily restoring the original representation.

Experimental framework

COMPUTATION



EXPERIMENTAL PROGRESSION

A Learn the task-sufficient core

Train projection-specific hard-concrete gates and the linear POS readout while pretrained encoder weights remain fixed.

Gates	TRAIN
Readout	TRAIN
Encoder weights	FIXED

The sparse configuration reaches 94.98% primary-run POS accuracy.

B Apply a permanent lesion

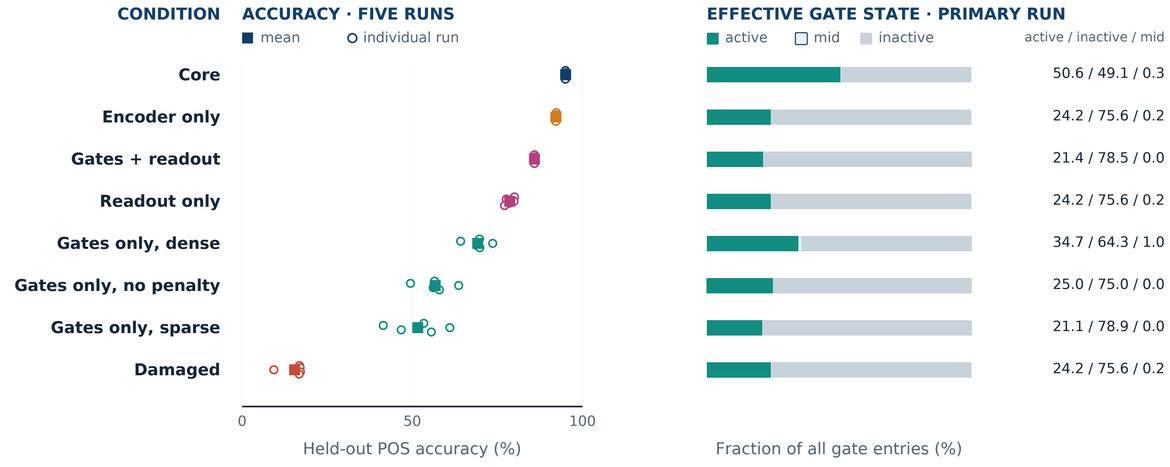
Gates used by the task-sufficient core are permanently disabled. Their effective values are zero and gradients are blocked.

C Recovery trainability

Condition	Gates	Readout	Encoder	Lesion
Gates only	TRAIN	FIXED	FIXED	OFF
Readout only	FIXED	TRAIN	FIXED	OFF
Gates + readout	TRAIN	TRAIN	FIXED	OFF
Encoder only	FIXED	FIXED	TRAIN	OFF

SETUP	bert-base-uncased | UD English-EWT | 17 POS tags | 25,147 development tokens
LESION	74 grouped Q/K/V input-coordinate masks; 5 attention-output, 9 FFN-intermediate, and 4 FFN-output projection-mask tensors; 37,248 entries clamped to -12 and gradient-blocked

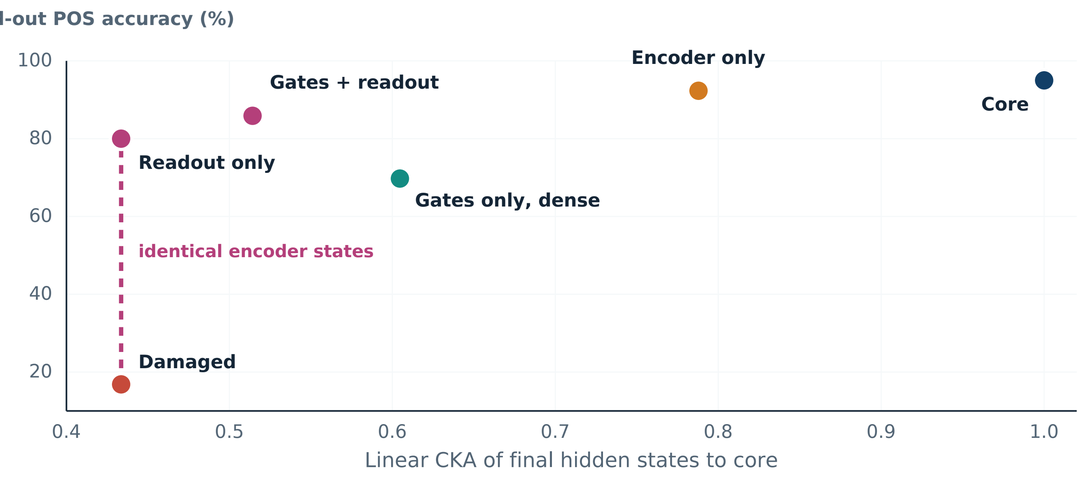
Performance and gate state



Circles are individual runs; squares are means. Gate state at the primary-run endpoint: active $p > 0.99$, inactive $p < 0.01$, otherwise intermediate. Lesioned entries are held at -12 and cannot change during recovery.

Performance vs. encoder geometry

Primary run

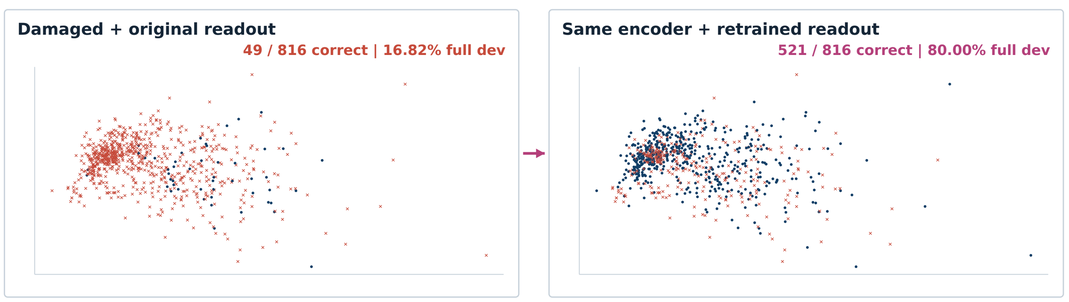


Linear CKA compares each condition's final hidden states with the pre-lesion core. Readout-only preserves the damaged hidden states (CKA 0.43) while primary-run accuracy rises from 16.82% to 80.00%.

Readout remapping exposes surviving POS information

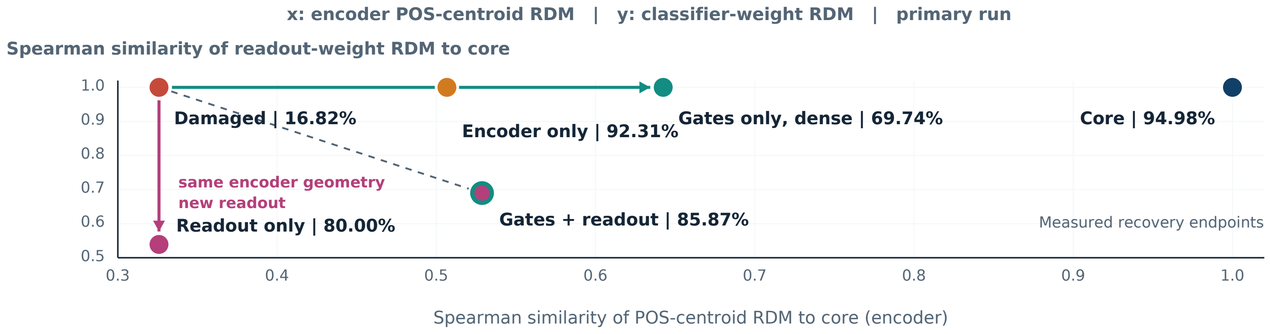
Readout-only recovery

Core-fitted PC1-PC2 view of the same 816 POS-balanced token states; classification uses all 768 dimensions



POS-centroid geometry vs. readout-weight geometry

x: encoder POS-centroid RDM | y: classifier-weight RDM | primary run

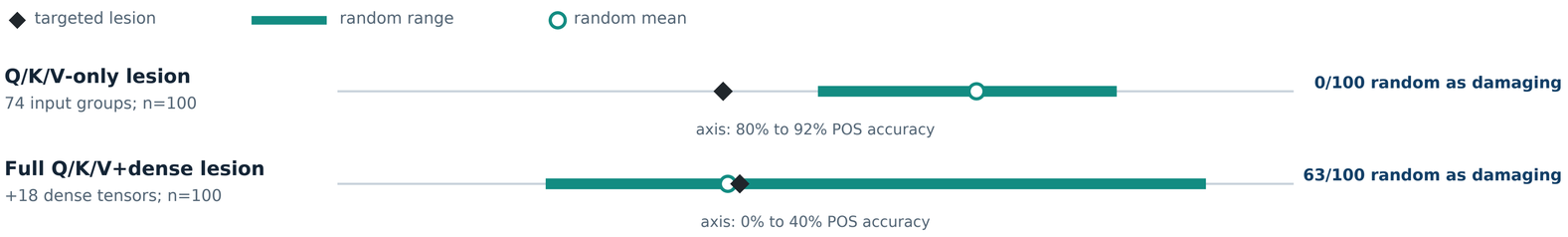


Recovery extends across POS classes

Primary run; recall within each gold POS class (%)

	NOUN	PUNCT	VERB	ADP	DET	ADJ	ADV	PRON	PROPN	CCONJ	AUX	NUM	PART	SCONJ	X	INTJ	SYM
Core (pre-lesion)	94	100	96	98	99	93	89	99	83	99	99	99	99	88	7	77	78
Damaged	98	0	0	1	0	0	0	0	3	0	1	0	0	0	0	0	0
Readout only	89	98	74	98	95	64	64	92	35	97	90	71	30	36	0	43	16
Gates + readout	90	99	81	88	97	73	75	96	55	97	89	93	97	48	0	53	62

Robustness and controls



- Q/K/V only: the targeted 74-group lesion was more damaging than all 100 size-matched random lesions.
- Full lesion: with the same Q/K/V and dense counts chosen at random, 63/100 lesions were at least as damaging.

Conclusions

Performance can recover without relearning the underlying transformations.

The extent to which the original geometry is recovered depends on the degrees of freedom of the recovery protocol.

Circuit necessity is conditional on the permitted rerouting strategy. A component may be necessary under the original fixed routing and readout, but cease to be so when the readout can be retrained or surviving transformations can be reselected.

LIMITATIONS

Our evaluations were based on a single task and a single backbone model.

EXPLORE THE EXPERIMENT

mechint25.github.io

References: Louizos et al. (2018); Cao et al. (2021). Data: Universal Dependencies English-EWT.